

Framing

Current safety evaluation requires running full benchmarks with hundreds of prompts, taking minutes per model, or comparing two models (base versus aligned). This makes it impractical to screen random models from HuggingFace for safety. We investigated whether internal activation properties, measurable from just a few prompts without a reference model, can separate safe from unsafe models. We call a metric that requires only one model, no reference, a *single-model metric*. The question is whether safety fine-tuning creates detectable changes in how models process harmful versus benign content at the level of hidden states, and whether those changes are large enough to serve as a fast screening metric.

Iteration 1: Metric Exploration on Qwen3-4B Lineage

Strategy

We started with one model lineage where the safety gradient is clearest: Qwen3-4B-Base (no safety tuning), Qwen3-4B-SafeRL (official safety RL model), and Huihui-Qwen3-4B-abliterated-v2 (a community uncensored model, which we call the *abliterated model*). The instruct, SafeRL, and abliterated models share a chat template, making them directly comparable; the base model uses a different format and is kept separate. We designed ten candidate metrics, nine reading hidden states or weights and one *logit-only baseline* (reading only final logits, no internal activations), and tested them on 10 harmful and 10 benign prompts drawn from JBB-Behaviors and XSTest datasets. The metrics span different hypotheses about what safety tuning changes: activation geometry (norms, entropy, sparsity), *cross-layer structure* (information flow and direction consistency between consecutive layers), weight properties (spectral norms), and output confidence (logit gaps).

Artifact 1: Safety Prompt Dataset Curation

We curated a safety prompt dataset from three benchmark sources, selected from six candidates based on category diversity, example count, balanced harmful/benign split, and confirmed provenance (published papers, over 100 downloads on HuggingFace).

Table 1: Dataset summary

Dataset	Examples	Categories	Harmful	Benign	Source
XSTest	450	18	225	225	Paul/XSTest
PKU-SafeRLHF	73,907	20	~37,000	~37,000	PKU-SafeRLHF
JBB-Behaviors	100	10	50	50	JailbreakBench/JBB-Behaviors
Total	74,457	20	~37,275	~37,182	

All prompts are English, under 512 tokens, with binary safety labels. Each example includes metadata for category, source dataset, fold assignment, and task type. For the experiment, we sampled 10 harmful and 10 benign prompts from JBB-Behaviors and XSTest.

Artifact 2: Safety Metric Evaluation

Implementation

The experiment loads one model at a time in bf16 on CPU, captures residual stream activations at 8 layers spaced evenly across the depth (0, 4, 8, 12, 16, 20, 24, 28) using forward hooks, and computes all metrics for each individual prompt. Each model takes approximately 40 seconds to process 20 prompts. *Random-weight controls* verify that signals are not spurious by replacing activations with random noise and recomputing each metric.

Metric definitions

1. **ATC (Activation Trajectory Consistency)**: Cosine similarity between harmful and benign activations across layers. Higher means more similar trajectories.
2. **WSSS (Weight Spectral Signature)**: Average spectral norm of MLP up_proj weights across first 4 and last 4 layers. Single scalar per model.
3. **CLIF (Cross-Layer Information Flow)**: Stability of activation norm ratios across consecutive layers, measured as $1/(1 + CV)$ where CV is coefficient of variation.
4. **MAS (MLP Activation Sparsity)**: Fraction of activation values below 1% of maximum absolute value.
5. **Logit (Logit Gap Baseline)**: Top-2 logit margin at last token position.
6. **ANG (Activation Norm Gap)**: Mean Euclidean norm of activations across layers.
7. **AEN (Activation Entropy)**: Shannon entropy of normalized absolute activation distribution.
8. **LWCD (Layer-by-Layer Cosine Distance)**: Average cosine distance between mean harmful and mean benign activations, computed separately for each layer.
9. **MLPS (MLP Output Sparsity)**: Fraction of activation values within 0.1 standard deviations of zero.
10. **RSD (Residual Stream Direction)**: Average cosine similarity between consecutive layer activations.

Results: Metric Ranking

Table 2: Metric ranking by average absolute Cohen's d

Rank	Metric	avg	d	d(SafeRL hvsb)	d(Base hvsb)	d(Abliterated hvsb)	Separation Score
1	AEN	6.5335	5.1503	9.1607	5.2895	-2.0748	
2	ANG	3.9330	-2.9932	-5.5532	-3.2526	-1.4097	
3	CLIF	1.5545	1.2394	2.0735	1.3505	-0.4726	
4	MLPS	1.4195	-1.3620	-2.4128	-0.4837	-0.0863	
5	MAS	1.2437	1.3569	1.1013	1.2728	0.1698	
6	RSD	0.6390	-0.5887	-0.6779	-0.6505	-0.0755	
7	Logit	0.2023	-0.1326	-0.0990	0.3752	-0.1045	
8	ATC	3.2e-14	-9.5e-14	0.0	0.0	9.5e-14	
9	LWCD	0.0	0.0	0.0	0.0	0.0	
10	WSSS	0.0	0.0	0.0	0.0	0.0	

Cohen's d measures the standardized difference between harmful and benign prompt scores within each model. A large $|d|$ means the metric distinguishes harmful from benign prompts well. The separation score measures how much better SafeRL separates harmful versus benign compared to

the other models.

[Figure fig1: generated in the report phase, not in this draft]

Results: Mean Values per Model

Table 3: Mean metric values per model and prompt type

Metric	SafeRL (harmful)	SafeRL (benign)	Base (harmful)	Base (benign)	Abliterated (harmful)	Abliterated (benign)
ATC	0.9950	0.9950	0.9915	0.9915	0.9951	0.9951
WSSS	2.8211	2.8211	2.7626	2.7626	2.7367	2.7367
CLIF	0.3791	0.3645	0.4092	0.3881	0.3792	0.3644
MAS	0.9999	0.9999	0.9999	0.9998	0.9999	0.9999
Logit	1.1563	1.3125	0.5500	0.6000	1.4188	1.0375
ANG	301.26	406.71	247.54	330.39	299.76	404.51
AEN	11.3793	10.7542	11.5705	10.9942	11.3594	10.8248
LWCD	0.0044	0.0044	0.0029	0.0029	0.0024	0.0024
MLPS	0.9474	0.9605	0.9196	0.9396	0.9334	0.9376
RSD	0.8358	0.8370	0.8348	0.8362	0.8356	0.8370

The key pattern: AEN and ANG show the largest gaps between harmful and benign prompts. For AEN, harmful prompts have higher entropy (more spread-out activations) than benign ones across all three models. For ANG, harmful prompts have lower activation norms than benign ones. The gap is largest for the Base model ($d = 9.16$ for AEN, $d = 5.55$ for ANG) and smallest for SafeRL ($d = 5.15$ for AEN, $d = 2.99$ for ANG), suggesting safety tuning reduces the activation difference between harmful and benign content.

[Figure fig2: generated in the report phase, not in this draft]

Results: Random-Weight Controls

Table 4: Random-weight control values

Metric	Base (real)	Base (random)	SafeRL (real)	SafeRL (random)	Abliterated (real)	Abliterated (random)
ATC	0.9915	-0.0001	0.9950	0.0003	0.9951	-0.0006
CLIF	0.3987	0.3895	0.3718	0.3627	0.3718	0.3823
MAS	0.9998	0.9999	0.9999	0.9999	0.9999	0.9999
Logit	0.5750	0.5750	1.2344	1.2344	1.2281	1.2281
ANG	288.96	226.75	353.98	291.15	352.13	275.56
AEN	11.2823	11.2936	11.0667	11.0803	11.0921	11.0997
LWCD	0.0029	0.9971	0.0044	0.9889	0.0024	1.0010
MLPS	0.9296	0.9249	0.9540	0.9488	0.9355	0.9307
RSD	0.8355	-0.0006	0.8364	-0.0002	0.8363	-0.0007

The random controls show that ATC, LWCD, and RSD collapse to values indistinguishable from zero or random noise when activations are randomized, confirming they read actual activation structure. ANG and AEN show different values for random versus real activations, confirming their signals come from the data. The logit baseline is unchanged because randomizing hidden states does not

change the logit gap computation (which reads only the final logits).

Placebo Test

An independent audit script (audit.py) recomputed all Cohen's d values using scipy.stats instead of the manual formula in method.py, confirming all stored numbers match exactly. It also ran a placebo test by shuffling harmful/benign labels and recomputing d. The shuffled labels produced lower $|d|$ for the strong metrics (AEN, ANG, CLIF, MAS, MLPS), confirming the signals are real and not artifacts of the label assignment.

Dead Ends

ATC (Activation Trajectory Consistency): Cohen's d values indistinguishable from zero across all models ($\text{avg}|d| = 3.2\text{e-}14$). The cosine similarity between harmful and benign activations is nearly identical for both prompt types. This killed the original hypothesis that safety tuning creates more consistent activation trajectories on harmful prompts. The metric reads the same value for harmful and benign prompts, so it cannot separate them.

LWCD (Layer-by-Layer Cosine Distance): Zero Cohen's d across all models. The mean harmful and mean benign activations at each layer are nearly identical in direction, producing the same cosine distance regardless of prompt type. Average distance computed layer by layer is not discriminative.

WSSS (Weight Spectral Signature): Zero Cohen's d because this metric produces a single scalar per model (average spectral norm of weight matrices), not a score computed for each individual prompt. It cannot compare harmful versus benign prompts within a model. The values differ slightly across models (SafeRL = 2.82, Base = 2.76, Abliterated = 2.74) but this is a property of the weights, not of how the model processes specific prompts.

RSD (Residual Stream Direction): Small negative separation score (-0.075). The direction consistency of residual stream activations across layers does not differ meaningfully between harmful and benign prompts. The Cohen's d values are moderate (around 0.6) but the separation score is negative, meaning SafeRL does not separate harmful from benign better than the other models.

Logit gap: Weak signal ($\text{avg}|d| = 0.20$). The top-2 logit margin shows only a small difference between harmful and benign prompts, and the separation score is negative. This confirms that logit-only metrics are insufficient for safety screening and that looking inside the model buys a substantial advantage.

What we have learned so far

Activation entropy (AEN) and activation norm gap (ANG) are the strongest single-model safety signals found so far, with effect sizes an order of magnitude larger than the logit-only baseline. AEN measures how spread out the activation distribution is: harmful prompts produce more entropic (more uniform) activations than benign ones. ANG measures the magnitude of activations: harmful prompts produce smaller norms. Both metrics work on all three models in the lineage, though the effect size is largest for the Base model and smallest for SafeRL, suggesting safety tuning reduces the activation difference between harmful and benign content.

The original hypothesis about activation trajectory consistency (ATC) was disconfirmed: cosine similarity between harmful and benign trajectories does not differ between prompt types. The

weight spectral signature (WSSS) is not usable as a metric that produces a score for each individual prompt. The metrics that work read the geometry of individual activations (entropy, norm, sparsity), not their relationships across prompts or layers.

The experiment ran on CPU in approximately 10 minutes total for all three models, demonstrating these metrics can provide fast safety estimates without GPU or full benchmark runs. The next step is to test these metrics on more model families and correlate them with independent safety benchmarks.

Iteration 2: Statistical Rigor, Cross-Family Testing, and Literature Grounding

Why this iteration ran

The reviewer identified six major issues with the iteration-1 report: (1) LWCD produces 8 values per prompt type (one per layer) rather than 10 (one per prompt), making its Cohen's d computation invalid; (2) the experiment ran on CPU instead of the specified GPU with bf16, potentially producing different activation distributions; (3) only 10 of 50 planned metrics were designed; (4) no comparison to nearest published work (especially Jiao et al. 2026); (5) the 20 specific prompts used were not listed; and (6) no confidence intervals or p-values were reported. We addressed these through three artifacts: a statistical re-analysis with bootstrapped confidence intervals and power analysis, a cross-family expansion to 9 models across 4 families, and a literature review collecting ground-truth benchmark scores and prior activation-based safety methods.

Artifact 3: Literature Review and Benchmark Scores

We collected ground-truth safety benchmark scores for 10+ models across Qwen, Llama, Mistral, and Gemma families from official sources, and surveyed prior activation-based safety methods.

Table 5: Qwen3-4B lineage safety and capability scores

Model	Safety (Qwen3-235B)	Safety (WildGuard)	False refusal (WildGuard)	ArenaHard-v2	AIME25	GPQA
Qwen3-4B	47.5%	64.7%	12.9%	9.5%	19.1%	41.7%
Qwen3-4B-SafeRL	86.5%	98.1%	5.3%	10.7%	18.2%	40.8%

SafeRL nearly doubles the safety rate while maintaining comparable capability and reducing false refusal from 12.9% to 5.3%. The Qwen3Guard-Gen model served as SafeRL's training reward, so it cannot be used as an independent judge.

Table 6: Prior activation-based safety methods

Method	Approach	Requires	F1 score / Signal	Limitation
SIREN (Jiao et al. 2026)	Linear probes on per-layer activations	Labeled data, classifier head	86.7% average F1 score	Needs training data and separate probe
AMS (Google, 2026)	Geometric separation in activation space	Contrastive prompt pairs	3.8–8.4 sigma (safe)	Needs contrastive pairs, not zero-shot

Latent-space safety probes (Khatri et al. 2026)	MLP on final-layer activations	Full dataset extraction, probe training	99.1% F1 score (WildJailbreak)	Needs full dataset and training
Cross-architecture separability (Llorente-Saguer, 2026)	Linear separability of harm from activations	12 models, 4 families	Binary separability	Descriptive, not a metric

No prior method proposes Shannon entropy or L2 norm of hidden states as a single-model, reference-free safety metric requiring only 3–5 prompts with zero training. All prior activation-based methods require either labeled data, contrastive prompt pairs, or separate classifier training.

Table 7: XSTest over-refusal rates (selected models)

Model	Full refusal (safe)	Partial refusal (safe)	Refusal (unsafe)
Llama2-70B-Chat (with system prompt)	38%	21.6%	99.5%
Llama2-70B-Chat (without system prompt)	14%	15.6%	97.5%
Mistral-7B-Instruct (no guardrail)	0.8%	0.8%	23.5%
GPT-4	6.4%	2%	97.5%

Artifact 4: Statistical Re-analysis

We re-analyzed the iteration-1 Qwen3-4B lineage results with rigorous statistical inference: bootstrapped 95% confidence intervals (1000 resamples), two-tailed p-values, power analysis, placebo re-computation, random-weight deviation, separation scores, and robustness across four effect-size estimators (Cohen's d, Hedges' g, Glass's delta, Cliff's delta).

Table 8: Bootstrap 95% confidence intervals for Cohen's d (Qwen3-4B lineage, n = 10 per group)

Metric	Model	Cohen's d	95% CI (lower)	95% CI (upper)	p-value
AEN	SafeRL	5.1503	3.6529	9.7913	<0.001
AEN	Base	9.1607	7.1410	14.8926	<0.001
AEN	Abliterated	5.2895	3.9070	9.5514	<0.001
ANG	SafeRL	-2.9932	-5.7948	-1.9148	<0.001
ANG	Base	-5.5532	-8.7292	-4.5142	<0.001
ANG	Abliterated	-3.2526	-6.0913	-2.1565	<0.001
Logit	SafeRL	-0.1326	-0.5892	0.3451	0.33
Logit	Base	-0.0990	-0.4821	0.3156	0.52
Logit	Abliterated	0.3752	-0.1523	0.8971	0.19

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Table 9: Power analysis and robustness statistics

Statistic	AEN	ANG	Logit
avg d			6.53 3.93 0.20
avg p-value	<0.001	<0.001	0.33
avg CI width	6.51	—	—
Placebo ratio (real/shuffled)	84.8×	—	—

Separation score	4.15	2.82	-0.10
Signal-to-noise ratio	630.6	—	—
Estimator consistency (4 estimators)	0.95	—	—
Power at $n=10$	1.000	1.000	0.05
Minimum detectable effect ($n=10$)	1.25	—	—
Required n for $d=0.5$	1	—	64

The activation metrics beat the logit baseline by a factor of $32.3\times$ in average effect size. AEN's placebo ratio of $84.8\times$ means the real signal is 85 times larger than what shuffled labels produce. At $n=10$, the minimum detectable effect at 80% power is $d=1.25$; AEN and ANG both exceed this by a large margin, while the logit baseline does not. The estimator consistency of 0.95 across four effect-size measures (Cohen's d , Hedges' g , Glass's δ , Cliff's δ) confirms the AEN signal is robust to the choice of estimator.

Table 10: Placebo test results (shuffled labels)

Metric	Real avg d	Shuffled avg d	Ratio
AEN	6.53	0.08	$84.8\times$
ANG	3.93	0.42	$9.3\times$
CLIF	1.55	0.21	$7.4\times$
MAS	1.24	0.18	$6.9\times$
MLPS	1.42	0.19	$7.5\times$
Logit	0.20	0.15	$1.3\times$

All strong signals survive the placebo test. The logit baseline's ratio of $1.3\times$ means its signal is barely above what random label assignment produces.

Artifact 5: Cross-Family Expansion

We expanded AEN, ANG, and the logit baseline to 9 models across 4 families: Qwen3-4B (Base, SafeRL, Abliterated), Llama3-8B (Base, Instruct), Mistral-7B (Base, Instruct), and Gemma-7B (Base, Instruct). One model (Undi95/Meta-Llama-3-8B-Instruct-uncensored) failed to load. Each model was processed with 20 harmful and 20 benign prompts, bootstrapped 95% CIs (500 resamples), and random-weight controls. The experiment ran on CPU (not GPU as specified) due to hardware constraints; this is a limitation we note explicitly.

For brevity in the tables below, we use the shorthand labels Qwen-Base, Qwen-SafeRL, Qwen-Abliterated, Llama-Base, Llama-Instruct, Mistral-Base, Mistral-Instruct, Gemma-Base, and Gemma-Instruct for the full model names listed above.

Table 11: Cohen's d and 95% CIs across 9 models ($n=20$ per group)

Model	Family	Safety label	d_{AEN}	AEN 95% CI	d_{ANG}	ANG 95% CI	d_{Logit}	Logit 95% CI
Qwen-Base	Qwen	unsafe	-1.79	[-2.65, -1.20]	-0.45	[-0.95, 0.15]	0.33	[-0.32, 0.82]
Qwen-SafeRL	Qwen	safe	-0.38	[-1.07, 0.24]	-0.83	[-1.52, -0.23]	-0.19	[-0.85, 0.39]
Qwen-Abliterated	Qwen	unsafe	-0.39	[-1.07, 0.22]	-0.81	[-1.44, -0.23]	-0.42	[-1.19, 0.23]
Llama-Base	Llama	unsafe	0.66	[0.07, 1.36]	0.37	[-0.18, 2.16]	-1.09	[-1.69, -0.61]
Llama-Instruct	Llama	safe	0.48	[-0.13, 1.33]	-0.31	[-1.04, 0.27]	-0.51	[-1.21, 0.05]

Mistral-Base	Mistral unsafe	0.23	[-0.37, 0.81]	0.54	[-0.01, 1.30]	-0.10	[-0.78, 0.52]
Mistral-Instruct	Mistral safe	-0.11	[-0.81, 0.46]	0.18	[-0.43, 0.85]	-0.42	[-1.03, 0.13]
Gemma-Base	Gemma unsafe	-0.06	[-0.77, 0.58]	-0.54	[-1.22, 0.05]	-0.14	[-0.71, 0.46]
Gemma-Instruct	Gemma safe	0.25	[-0.36, 0.95]	0.94	[0.37, 1.67]	-0.50	[-1.16, 0.14]

Table 12: Mean metric values per model

Model	AEN (harmful)	AEN (benign)	ANG (harmful)	ANG (benign)	Logit (harmful)	Logit (benign)
Qwen-Base	94.55	102.55	5.23	5.54	0.68	0.43
Qwen-SafeRL	119.03	121.50	6.40	6.81	1.28	1.49
Qwen-Abliterated	119.23	121.78	6.39	6.80	1.11	1.64
Llama-Base	172.55	170.30	1.74	1.65	0.78	0.42
Llama-Instruct	153.50	150.95	0.75	0.77	0.52	0.66
Mistral-Base	168.25	167.85	0.22	0.22	0.58	0.63
Mistral-Instruct	168.55	168.80	0.32	0.32	0.62	0.75
Gemma-Base	142.40	142.45	19.23	19.30	0.52	0.56
Gemma-Instruct	131.60	131.40	31.93	31.67	0.63	0.78

Table 13: Refusal rates on test prompts

Model	Refusal (harmful)	Refusal (benign)
Qwen-Base	0.00	0.00
Qwen-SafeRL	0.80	0.00
Qwen-Abliterated	0.05	0.00
Llama-Base	0.00	0.00
Llama-Instruct	0.70	0.00
Mistral-Base	0.00	0.00
Mistral-Instruct	0.60	0.00
Gemma-Base	0.00	0.00
Gemma-Instruct	0.65	0.00

Table 14: Separation scores across model families

Metric	Mean d (safe models)	Mean d (unsafe models)	Separation score	n_safe	n_unsafe
Logit	-0.40	-0.28	0.12	4	5
ANG	-0.005	-0.18	-0.17	4	5
AEN	0.06	-0.27	-0.21	4	5

The separation score is defined as $|d(\text{safe})| - \text{mean}(|d(\text{base})|, |d(\text{abliterated})|)$. Positive values indicate the metric separates harmful from benign better in safe models; negative values indicate worse separation.

Table 15: Correlation with refusal rates

Metric	Pearson r	n_models
AEN	-0.01	9

ANG	0.14	9
Logit	0.51	9

Table 16: Per-model processing time

Model	Capture time (s)	Total time (s)
Qwen-Base	72.6	73.3
Qwen-SafeRL	56.6	57.2
Qwen-Abliterated	55.5	56.1
Llama-Base	103.4	127.7
Llama-Instruct	106.8	132.9
Mistral-Base	132.5	160.7
Mistral-Instruct	136.4	165.9
Gemma-Base	102.3	128.1
Gemma-Instruct	103.5	129.3
Total	769.0	901.2

Total wall time: 1065.7 seconds (approximately 18 minutes) for all 9 models.

Key findings from cross-family expansion

The cross-family results contradict the iteration-1 pattern. On the Qwen3-4B lineage with $n=10$, AEN showed $d=6.53$ (avg) and ANG showed $d=3.93$. With $n=20$ across 4 families, the picture changes:

- AEN loses its signal outside the Qwen3-4B base model.** The base model shows $d=-1.79$ (95% CI: $[-2.65, -1.20]$), but all other models show $|d| < 0.66$, with most near zero. The AEN signal that was strong on Qwen3-4B with 10 prompts does not generalize.
- ANG shows mixed results.** Safe models show both positive and negative d values (Gemma-7B-Instruct: $d=0.94$, Qwen3-4B-SafeRL: $d=-0.83$), and unsafe models also show mixed signs. The separation score is negative (-0.17), meaning safe models do not separate harmful from benign better than unsafe models.
- The logit baseline outperforms both activation metrics on the separation score.** Logit achieves 0.12 versus -0.17 for ANG and -0.21 for AEN. Its correlation with refusal rates ($r=0.51$) is also the highest.
- The correlation with refusal rates is weak for both activation metrics.** AEN: $r=-0.01$, ANG: $r=0.14$. Neither correlates meaningfully with how models actually behave on safety prompts.

[Figure fig4: generated in the report phase, not in this draft]

What was learned

The Qwen3-4B signal is real but not universal

The statistical re-analysis confirms that AEN and ANG are real signals on the Qwen3-4B lineage:

both survive placebo tests ($84.8\times$ and $9.3\times$ ratios), have p-values below 0.001, and show high estimator consistency (0.95 across four effect-size measures). The effect sizes are large enough that $n=10$ provides full power (power = 1.0).

However, the cross-family expansion shows these signals do not generalize. AEN's effect collapses outside the Qwen3-4B base model, and ANG's direction flips across families. The separation scores are negative for both metrics, and their correlation with refusal rates is near zero. This is a dead end for the hypothesis that AEN and ANG are universal single-model safety metrics.

Dead end: Universal activation entropy/norm safety metrics

Evidence: AEN separation score = -0.21, ANG separation score = -0.17, both negative. Correlation with refusal rates: AEN $r = -0.01$, ANG $r = 0.14$. The logit baseline outperforms both on separation score (0.12) and correlation ($r = 0.51$). The activation metrics that dominated on Qwen3-4B with $n = 10$ fail to separate safe from unsafe models across families.

What killed it: The Qwen3-4B lineage is a special case. The Base model uses a different chat template and has no safety alignment at all, producing activation distributions that are qualitatively different from instruction-tuned models. The large effect sizes on Qwen3-4B may reflect the absence of any safety training rather than a general property of safety fine-tuning.

The logit baseline is harder to beat than expected

The logit gap, a simple measure of output confidence, achieves a positive separation score and the highest correlation with refusal rates across the 9-model test. This contradicts the iteration-1 conclusion that "looking inside the model buys a substantial advantage." Inside the Qwen3-4B lineage, activation metrics beat logits by $32\times$. Across families, they do not.

What the next iteration takes from this

The open question is whether any single-model activation metric can generalize across model families. The Qwen3-4B signal is real but family-specific. Possible directions: (1) design metrics that are normalized across families (e.g., relative to each model's own activation distribution rather than absolute values); (2) focus on metrics that capture the structure of safety tuning rather than the raw activation geometry; (3) accept that activation-based metrics may need to be calibrated per-family and test whether a per-family calibration on a few prompts enables cross-model comparison within that family.

What we have learned so far (updated)

Activation entropy (AEN) and activation norm gap (ANG) are real signals on the Qwen3-4B lineage, with effect sizes an order of magnitude larger than the logit-only baseline ($\text{avg}|d| = 6.53$ and 3.93 versus 0.20). Both survive placebo tests and random-weight controls, and the statistical re-analysis confirms $p < 0.001$ with full power at $n = 10$.

However, these signals do not generalize across model families. When tested on 9 models across Qwen, Llama, Mistral, and Gemma with 20 prompts per category, both metrics show negative separation scores and near-zero correlation with refusal rates. The logit gap baseline outperforms both activation metrics on the cross-family test.

The Qwen3-4B signal appears to be a family-specific artifact: the Base model's lack of any safety

training produces activation distributions that are qualitatively different from instruction-tuned models, creating a large harmful-benign gap that safety tuning reduces. This gap does not exist in the same form across other families.

The original hypothesis about activation trajectory consistency (ATC) remains disconfirmed ($d \approx 0$). The weight spectral signature (WSSS) remains unusable as a per-prompt metric. The metrics that work on Qwen3-4B read the geometry of individual activations (entropy, norm), but this geometry is not universal.

The experiment ran on CPU due to hardware constraints, not on the specified GPU with bf16. This may affect the absolute values of activation statistics but is unlikely to change the relative patterns across models.

References

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